polynomial˙report

Polynomial

$$f(q) = 3q^3 + 7q^2 + 2q + 1$$

Naive

$$g_{\text{naive}}(X) = 3X^3 + 7X^2 + 2X + 1$$

$$\mathbb{E}[g_{\text{naive}}(X)] = \frac{\Delta}{\epsilon}^{2} (18q + 14) + 3q^3 + 7q^2 + 2q + 1$$

$$\begin{aligned}\operatorname{Var}(g_{\text{naive}}(X)) = \frac{\Delta}{\epsilon}^{2} \Bigg(6480\frac{\Delta}{\epsilon}^{4} + 2\frac{\Delta}{\epsilon}^{2} \left(1458q^2 + 2268q + 634\right) + 162q^4 + 504q^3 \\ + 464q^2 + 112q + 8\Bigg)\end{aligned}$$

$$\begin{aligned}\operatorname{MSE}(g_{\text{naive}}) = \frac{\Delta}{\epsilon}^{2} \Bigg(6480\frac{\Delta}{\epsilon}^{4} + 2\frac{\Delta}{\epsilon}^{2} \left(1620q^2 + 2520q + 732\right) + 162q^4 + 504q^3 \\ + 464q^2 + 112q + 8\Bigg)\end{aligned}$$

$$\operatorname{Bias}(g_{\text{naive}}) = \frac{\Delta}{\epsilon}^{2} (18q + 14)$$

Unbiased

$$g_{\text{unb}}(X) = 3X^3 + 7X^2 + 2X + \frac{\Delta}{\epsilon}^{2} (-18X - 14) + 1$$

$$\mathbb{E}[g_{\text{unb}}(X)] = 3q^3 + 7q^2 + 2q + 1$$

$$\begin{aligned}\operatorname{Var}(g_{\text{unb}}(X)) = \frac{\Delta}{\epsilon}^{2} \Bigg(4536\frac{\Delta}{\epsilon}^{4} + 2\frac{\Delta}{\epsilon}^{2} \left(1134q^2 + 1764q + 562\right) + 162q^4 + 504q^3 \\ + 464q^2 + 112q + 8\Bigg)\end{aligned}$$

$$\begin{aligned}\operatorname{MSE}(g_{\text{unb}}) = \frac{\Delta}{\epsilon}^{2} \Bigg(4536\frac{\Delta}{\epsilon}^{4} + 2\frac{\Delta}{\epsilon}^{2} \left(1134q^2 + 1764q + 562\right) + 162q^4 + 504q^3 \\ + 464q^2 + 112q + 8\Bigg)\end{aligned}$$

Comparison

$$\text{mean gap} = \frac{\Delta}{\epsilon}^2 \left(-18q - 14\right)$$

$$\text{variance gap} = \frac{\Delta}{\epsilon}^4 \left(-1944\frac{\Delta}{\epsilon}^2 - 648q^2 - 1008q - 144\right)$$

$$\text{MSE gap} = \frac{\Delta}{\epsilon}^4 \left(-1944\frac{\Delta}{\epsilon}^2 - 972q^2 - 1512q - 340\right)$$

$$\text{Bias}(g_{\text{naive}})^2 = \frac{\Delta}{\epsilon}^4 \left(324q^2 + 504q + 196\right)$$